Rotational linearization obstructions in a doubled Einstein–scalar shell model

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Abstract

We work out a rotational linearization obstruction for two Einstein–scalar regions joined across a timelike shell whose action is a scalar-dependent tension. The integrated momentum constraint and mixed Israel equation give an exact functional containing the shell's area–boost-angle term. At a rotation-invariant background its first variation vanishes for arbitrary matching variations, including variations of the shell geometry. Consequently a nonzero quadratic rotational functional cannot be cancelled by an unrestricted second-order correction. In a five-dimensional radial background, an explicit scalar master reconstruction and finite propagation produce smooth compactly supported linear solutions before shell contact. We exhibit an exact background with smooth interiors, including its fixed potential and shell tension, on which a rotating quadrupole has a strictly nonzero obstruction. For a fixed-profile quadrupole family, the six rotational conditions reduce to commutation of two trace-free symmetric matrices. We describe the regular and singular geometry of this finite condition, including the exact criterion for a linearized matrix direction to integrate to a commuting curve. The Noether and linearization-instability mechanisms are established; the result is an explicit physical-interface application with a computable witness. The matrix criterion is not a sufficiency theorem for nonlinear spacetimes.

Keywords: Einstein–scalar equations; timelike shells; linearization instability; gravitational constraints; momentum maps.

1 Question, contribution and relation to prior work

A solution of the linearized field equations need not be the first derivative of a family of exact solutions. At a symmetric background, the second derivative of the constraints can impose a quadratic condition on a linear solution. This is the classical linearization-stability problem [1–3]. For an internal timelike shell, applying this principle requires accounting for the cut shared by the two bulk Cauchy sections. Compact support of the *first* variation does not remove an arbitrary *second* variation at that cut.

This paper gives a direct calculation for a pure-potential scalar shell. Its main steps are the exact cut identity (6), the vanishing of the first derivative on all matching corrections (9), and the explicit nonzero value (22). The witness exists on a short slab before contact with the shell, so the central result uses a standard hyperbolic Cauchy theorem rather than a shell initial-boundary existence theorem. The finite-dimensional calculation in Section 6 distinguishes passing the rotational tests from lying in a regular part of their zero set.

The general construction is not new. Iyer–Wald supplies the bulk Noether identity [4]. Harlow–Wu's completed boundary Hamiltonian already has the combination of Noether charge, boundary Lagrangian and corner potential evaluated in Appendix B [5]. The physical-shell Hamiltonian of Hájíček–Kijowski, with its erratum, supplies direct precedent for induced matching and the area–boost-angle pair [6, 7]. Recent general treatments explicitly relate second-order fields, boundary terms and quadratic symmetry constraints [8, 9]; the compact-Cauchy Taub mechanism is also displayed directly in [10]. Our calculation specializes these mechanisms to a two-sided scalar/tension interface and evaluates a smooth obstructed family. It does not provide a new general Noether formalism or a classification of the nonlinear constraint range. The accompanying primary-source comparison records the exact passages used and the remaining historical-source access gaps.

2 Model, matching and regularity

The interface identity below holds in spacetime dimension $d+1$, $d \geq 2$. Use signature $(-+\cdots+)$ and $8\pi G = 1$. Two bulk regions M_+ and M_- meet at a common timelike shell W. Their outward spacelike unit normals are n_i, and $K_{i,ab} = h_a{}^A h_b{}^B \nabla_A n_{i,B}$. The action is

$$S = \sum_{i=\pm} \int_{M_i} \epsilon_i \left(\frac{R_i}{2} - \frac{(\nabla\phi_i)^2}{2} - U(\phi_i) \right) + \int_W \epsilon_W \left[K_+ + K_- - \sigma(\varphi) \right]. \tag{1}$$

There are two outward Gibbons–Hawking–York terms and one tension term. The smooth functions U and σ are fixed. Exact induced matching is part of the configuration space:

$$X_+^* g_+ = X_-^* g_- = h, \qquad X_+^* \phi_+ = X_-^* \phi_- = \varphi. \tag{2}$$

Writing $p_i = n_i \phi_i$, the metric and scalar junction residuals are

$$I_{ab} = \sum_i (K_{i,ab} - K_i h_{ab}) + \sigma h_{ab}, \qquad P = p_+ + p_- + \sigma'(\varphi). \tag{3}$$

The equations set both residuals to zero. In particular, scalar-dependent tension has no intrinsic derivative kinetic term and its stress is proportional to h_{ab}.

Let each spatial section C_i be compact with common closed boundary S, and assume no external spatial boundary. The induced metric c on S is the same from both sides. A common reference vector J is tangent to S and is extended smoothly to each C_i. It is fixed when fields vary. For the explicit example, C_i are four-balls and J is a rotation.

All differentiations below take place in a local chart of exactly matched configurations. Moving embeddings and Cauchy sections are pulled back to fixed reference regions by smooth collars. A hypothetical exact family is assumed C^2 in its parameter in a piecewise C^k norm, $k \geq 4$, on a compact slab, with uniformly nondegenerate metrics and spacelike sections. These are sufficient classical assumptions for the stated products, traces and differentiations. No manifold theorem for the full solution space is assumed. The second variation may occupy every angular, metric, scalar and shell sector compatible with exact matching.

3 The rotational constraint with its shell cut

Let u_i be the future unit normal to C_i, s_i its outward unit normal to S, and τ the common future normal to S within W. Define the boost angles by

$$n_i = \cosh\eta_i \, s_i + \sinh\eta_i \, u_i, \qquad \tau = \cosh\eta_i \, u_i + \sinh\eta_i \, s_i. \tag{4}$$

For the spatial metric γ_i and extrinsic curvature k_i, use the canonical densities

$$\pi_i^{ab} = \frac{\sqrt{\gamma_i}}{2}(k_i^{ab} - k_i\gamma_i^{ab}), \qquad \varpi_i = \sqrt{\gamma_i}\, u_i\phi_i, \qquad \mathcal{H}_{i,a} = -2D_b\pi_{i\,a}^{b} + \varpi_i D_a\phi_i. \tag{5}$$

Here $k_{ab} = \gamma_a{}^A\gamma_b{}^B\nabla_A u_B$; densities, rather than tensors with an additional volume element, are used in the bulk integrals.

Lemma 3.1 (Exact rotational cut identity)**.** *For the matched configurations above,*

$$\begin{aligned}\mathcal{C}_J &:= \sum_i \int_{C_i} \mathcal{H}_{i,a}J^a - \int_S \sqrt{c}\, I(\tau, J) \\ &= \sum_i \int_{C_i} (\pi_i^{ab}\mathcal{L}_J\gamma_{i,ab} + \varpi_i J\phi_i) + \int_S \sqrt{c}\, J(\eta_+ + \eta_-).\end{aligned} \tag{6}$$

Every exact constraint-and-junction datum has $\mathcal{C}_J = 0$.

Proof. Differentiate the orthonormal frame (4) along the cut-tangent field J. Normalization and orthogonality give

$$k_i(s_i, J) = -K_i(\tau, J) - J\eta_i. \tag{7}$$

Integration by parts of the momentum constraint gives the bulk canonical term and the boundary term $-\int_S \sqrt{c}\, k_i(s_i, J)$. Since $h(\tau, J) = 0$, the trace and tension terms in (3) have zero mixed projection; hence $I(\tau, J) = \sum_i K_i(\tau, J)$. Substitution of (7) proves (6). □

Write $\vartheta = \eta_+ + \eta_-$ and $\rho_S = \sqrt{c}$. Suppose the background canonical fields and cut data are all J-invariant:

$$\mathcal{L}_J\gamma_0 = \mathcal{L}_J\pi_0 = J\phi_0 = \mathcal{L}_J\varpi_0 = \mathcal{L}_J\rho_{S0} = J\vartheta_0 = 0. \tag{8}$$

Lie derivatives of $\pi_0, \varpi_0, \rho_{S0}$ are density Lie derivatives. These assumptions hold for the radial time-symmetric section used below.

Proposition 3.2 (No linear second-order cancellation)**.** *At a background satisfying* (8)*,*

$$D\mathcal{C}_J|_0[w] = 0 \quad \textit{for every matching variation } w. \tag{9}$$

For a first variation v, the Hessian is

$$\begin{aligned}\mathcal{T}_J(v) := D^2\mathcal{C}_J|_0[v, v] = 2\sum_i \int_{C_i} (\delta\pi_i^{ab}\mathcal{L}_J\delta\gamma_{i,ab} + \delta\varpi_i J\delta\phi_i) \\ + 2\int_S \delta\rho_S\, J\delta\vartheta.\end{aligned} \tag{10}$$

If v vanishes with the required first jets near the initial shell cut, the last integral is zero.

Proof. Differentiate (6). Terms in which the variation multiplies a Lie derivative of a background field vanish by (8). The remaining first-order terms are $\int \pi_0\mathcal{L}_J\delta\gamma$, $\int \varpi_0 J\delta\phi$ and $\int \rho_{S0}J\delta\vartheta$. Integrate them by parts. There is no exterior flux because J is tangent to the bulk cuts, and S is closed. Background density invariance sets each integral to zero. Differentiating a second time leaves exactly the cross terms (10); all terms linear in a second variation vanish by the same calculation. Because the first derivative is zero, the displayed Hessian is independent of the choice of matching chart. □

Theorem 3.3 (Necessary rotational continuation condition)**.** *Let v be the derivative of a classical parameter-C^2 family of exact solutions of (1)–(3) through a background satisfying (8). Then $\mathcal{T}_J(v) = 0$. In particular, a smooth linear solution for which this quantity is nonzero cannot be the first derivative of such a family, even when its second derivative is unrestricted in the bulk and on the shell.*

Proof. Write the exact family in a matching chart as $\Psi_\varepsilon = \Psi_0 + \varepsilon v + \frac{1}{2}\varepsilon^2 w + o(\varepsilon^2)$. Since $\mathcal{C}_J(\Psi_\varepsilon) = 0$, its second derivative is $D^2\mathcal{C}_J[v, v] + D\mathcal{C}_J[w] = 0$. Proposition 3.2 removes the second term for every w. □

The support hypothesis removes a quadratic cut contribution of v. It is not the reason the arbitrary second-order shell correction drops out: that follows separately from (9). Nor does $\mathcal{T}_J = 0$ prove that a correction w exists. The Hessian convention here is twice the quadratic Taylor coefficient.

4 An admissible five-dimensional witness before shell contact

Assume an actual smooth radial background of the equations exists on the complete doubled slab, with

$$g_0 = \Omega(s)^2\eta, \qquad \phi_0 = \Phi(s), \qquad s = R^2 - T^2, \qquad W : \ s = b^2. \tag{11}$$

The two copies agree, $\Omega > 0$, and $f = \Phi_s \neq 0$ on a neighborhood of the annulus used below. This is an explicit hypothesis about an actual background, not a claim that arbitrary functions Ω, Φ solve the nonlinear equations. A completely specified example is given in Section 4.1. The registered model motivating this calculation is documented separately [11]; its detuning and branch certification are not inputs here.

Set $\mathcal{E} = X^A\partial_A$ and $L = \Omega_s/\Omega + 2f_s/f$. In the region where $f \neq 0$, define

$$\mathscr{L}F = \Box_\eta F - 2L\mathcal{E}F - \left(8L + \frac{8}{3}sf^2\right)F. \tag{12}$$

This is normally hyperbolic with smooth lower-order coefficients. The reconstruction is

$$h_{AB} = 2F(s\eta_{AB} - 3X_AX_B), \qquad \chi = -\frac{3(\mathcal{E}+4)F}{2\Omega^2 f}. \tag{13}$$

Lemma 4.1 (Short-slab admissibility)**.** *Let the smooth data $q = F|_{T=0}$ and $p = F_T|_{T=0}$ have support in $r_1 \le R \le r_2$, where $0 < r_1 < r_2 < b$. For a sufficiently short two-sided time slab, (12) and (13) produce a smooth solution of the complete linearized bulk and shell equations. The perturbation vanishes near the shell and near the cone $s = 0$ throughout this slab.*

Proof. On $s > 0$, direct substitution using the background equations gives the operator identities

$$\begin{aligned} \delta\mathscr{E}_{AB} &= \Omega^{-2}(3X_AX_B - s\eta_{AB})\mathscr{L}F, \\ \delta\mathscr{S} &= -\frac{3}{2\Omega^4 f}(\mathcal{E}+6)\mathscr{L}F, \end{aligned} \tag{14}$$

where $\mathscr{E}_{AB} = R_{AB} - \phi_A\phi_B - \frac{2}{3}Ug_{AB}$ and $\mathscr{S} = \Box_g\phi - U'$; Appendix A supplies the component calculation. Extend the coefficients of the auxiliary scalar equation smoothly away from a compact neighborhood of the intended causal support, and apply the smooth Cauchy theorem and finite

propagation [12, Theorem 3.2.11]. The support is contained in $r_1 - |T| \le R \le r_2 + |T|$. Choose, in addition to the background-domain restriction,

$$0 < T_* < \min\{r_1/2, (b - r_2)/2\}. \tag{15}$$

Then the causal support lies in $R > |T|$ and $R < b$, separated from both the cone and the shell $R = \sqrt{b^2 + T^2}$. Shrink T_* further, if needed, so this support remains in the neighborhood where the actual background coefficients are smooth and $f \neq 0$. On this support the coefficients are those of the actual background and (14) applies. The reconstructed fields extend smoothly by zero to the remaining physical region. They vanish on an open shell neighborhood, so zero embedding variation satisfies every linearized matching and junction condition there. All initial bulk constraints follow from the bulk equations. The auxiliary coefficient extension has made no assertion about a nonlinear background outside the physical neighborhood. □

The first-order embedding and field jets vanish near the shell. Thus the pullback identifications for a candidate exact family can be chosen with zero first parameter jet near the entire initial Cauchy section, including the time jet needed for its canonical data. This preserves the explicit first variation below while allowing arbitrary second-order pullback terms. At $T = 0$, write $D = r\partial_r$. Direct ADM variation gives

$$\begin{aligned} \delta\gamma_{ij} &= 2q(r^2\delta_{ij} - 3x_i x_j), \quad & \delta\pi^{ij} &= -\frac{3p}{2\Omega}x_i x_j, \\ \delta\phi &= -\frac{3(D+4)q}{2\Omega^2 f}, & \delta\varpi &= -\frac{3\Omega(D+5)p}{2f}. \end{aligned} \tag{16}$$

The shift by five in the last formula follows from $[\partial_T, \mathcal{E}] = \partial_T$. Define

$$\mathcal{B}(q,p) = \int \mathrm{d}r\, \mathrm{d}\Omega_3 \left[m_r(D+4)q(D+5)p + n_r qp\right], \quad m_r = \frac{9r^3}{4\Omega f^2}, \quad n_r = \frac{6r^7}{\Omega}. \tag{17}$$

Use identical data on the two copies (or opposite signs on both fields of one copy). Substitution into (10) yields

$$\mathcal{T}_J(v) = 4\mathcal{B}(Jq, p). \tag{18}$$

Choose $Q \in C_c^\infty(I)$ nonzero, $I \Subset (0,b)$ in the regular annulus, a real harmonic H with $JH \neq 0$, and a nonzero real inverse-time constant ω. Put $q = QH$, $p = \omega QJH$. Then

$$\mathcal{T}_J(v) = 4\omega\mathcal{B}_{\mathrm{rad}}(Q,Q)\int_{S^3}(JH)^2\mathrm{d}\Omega_3. \tag{19}$$

No symmetry of the radial bilinear is assumed. Compact radial integration by parts gives

$$\mathcal{B}_{\mathrm{rad}}(Q,Q) = \int_I \left[m_r(DQ)^2 + \{m_r(2 + 9r^2L) + n_r\}Q^2\right]\mathrm{d}r. \tag{20}$$

If w_I is the length of I, let $m_* = \min_{\bar{I}}(m_r r^2) > 0$ and $M_* = \max_{\bar{I}} m_r(-2 - 9r^2L)_+$. The one-dimensional Poincaré inequality implies

$$\mathcal{B}_{\mathrm{rad}}(Q,Q) \ge \left(\frac{\pi^2 m_*}{w_I^2} - M_*\right)\int_I Q^2\mathrm{d}r > 0 \tag{21}$$

on a sufficiently thin annulus. This is positivity of one diagonal radial form, not positivity of a gravitational energy.

For $n = x/r$, take $H = n_1^2 - n_2^2$ and $J = x_1\partial_2 - x_2\partial_1$. Then $JH = -4n_1n_2$ and $\int_{S^3}(JH)^2 = 4\pi^2/3$, so

$$\boxed{\mathcal{T}_J(v) = \frac{16\pi^2}{3}\omega\mathcal{B}_{\rm rad}(Q,Q) \neq 0.} \tag{22}$$

Lemma 4.1 makes this an actual smooth linear solution. Theorem 3.3 excludes a classical parameter-C^2 nonlinear continuation with this tangent. Here an admissible pure-gauge variation is the infinitesimal joint diffeomorphism action on bulk fields and shell embeddings, generated by a smooth vector field with a local flow near the compact initial section. Such a variation is tangent to an exact diffeomorphic family; the nonzero obstruction therefore excludes it.

4.1 A fully specified background and witness

In a fixed length unit, set $z = 1 + s$ and choose

$$\Omega = z^{-2}, \qquad \Phi = \sqrt{6}\log z, \qquad U(\phi) = 84e^{2\phi/\sqrt{6}} - 36e^{3\phi/\sqrt{6}}. \tag{23}$$

These fields solve the bulk equations on $z > 0$. Indeed, writing $u = \log\Omega$, the conformal Ricci formula gives

$$\begin{aligned} R_{AB} &= 12(u_s^2 - u_{ss})X_AX_B + (-16u_s - 4su_{ss} - 12su_s^2)\eta_{AB} \\ &= \frac{24X_AX_B + (32 - 24s)\eta_{AB}}{z^2} = \Phi_A\Phi_B + \frac{2}{3}U(\Phi)g_{AB}, \end{aligned} \tag{24}$$

because $\Phi_s = \sqrt{6}/z$ and $U(\Phi) = (48 - 36s)z^2$. Independently,

$$\Box_g\Phi = \Omega^{-2}(4s\Phi_{ss} + 10\Phi_s + 12su_s\Phi_s) = \sqrt{6}(10 - 18s)z^2 = U'(\Phi). \tag{25}$$

Join two identical interiors $s \le b^2$ at $b = 1/2$. Their outward normal is $n^A = X^A/(b\Omega_b)$ and their extrinsic curvatures and scalar normal derivatives are

$$K_{i,ab} = a_b h_{ab}, \quad a_b = \frac{(1 - 3b^2)(1 + b^2)}{b} = \frac{5}{8}, \qquad p_b = 2\sqrt{6}\,b(1 + b^2) = \frac{5\sqrt{6}}{4}. \tag{26}$$

With $\phi_b = \sqrt{6}\log(5/4)$, the fixed affine tension

$$\sigma(\phi) = \frac{15}{4} - \frac{5\sqrt{6}}{2}(\phi - \phi_b) \tag{27}$$

satisfies $\sigma(\phi_b) = 6a_b$ and $\sigma'(\phi_b) = -2p_b$. Thus both residuals in (3) vanish exactly. Induced matching is automatic. On $|T| \le 1/2$, the interior has $-1/4 \le s \le 1/4$, so the Cartesian fields are smooth through the center and cone within each interior, with $\Omega > 0$ and $f \neq 0$. Across the material shell the metric is piecewise smooth, with the extrinsic-curvature jump fixed by the junction equations. The initial section is time symmetric and rotation invariant.

For this background,

$$L = -\frac{4}{1 + s}, \qquad m_r = \frac{3}{8}r^3(1 + r^2)^4, \qquad n_r = 6r^7(1 + r^2)^2, \tag{28}$$

and the zeroth-order radial coefficient is

$$m_r(2 + 9r^2L) + n_r = m_r\frac{2 - 32r^2 - 18r^4}{(1 + r^2)^2} > 0 \quad (0 < r \le 1/5). \tag{29}$$

The numerator decreases on this interval and equals 432/625 at its upper endpoint. Hence any nonzero $Q \in C_c^\infty((1/10, 1/5))$ has $\mathcal{B}_{\mathrm{rad}}(Q, Q) > 0$, without the thin-annulus estimate. For example, take $Q = \exp[-1/((r - 11/100)(19/100 - r))]$ between 11/100 and 19/100, and zero elsewhere. Take $T_* = 1/32$ and $\omega = 1$ in the fixed units. Together with the displayed H and J, these choices specify a smooth linear state with (22) strictly positive on a fully specified exact background.

This illustrative potential and tension differ from the registered HDBLAST model. The example establishes that the present hypotheses are realized; it makes no priority claim for the exact background and no claim about its long-time stability or cosmological interpretation.

5 Quadrupole charges and commuting matrices

Keep Q, ω and the copy parity fixed, with $\mathcal{B}_{\mathrm{rad}}(Q, Q) > 0$. Every real degree-two harmonic on S^3 is $H_A(n) = n^T A n$ with $A \in V := \mathrm{Sym}_0(4, \mathbb{R})$. The family

$$q = Q H_A, \qquad p = \omega Q H_B, \qquad (A, B) \in V \times V \tag{30}$$

has dimension eighteen. For $M^T = -M$ and $J_M = (Mn) \cdot \partial_n$, the exact sphere moments give

$$\int_{S^3} H_A H_B = \frac{\mathrm{vol}(S^3)}{12} \operatorname{tr}(AB), \qquad J_M H_A = H_{[A,M]}. \tag{31}$$

Consequently

$$\mathcal{T}_{J_M}(v) = \frac{\mathrm{vol}(S^3)}{3} \omega \mathcal{B}_{\mathrm{rad}}(Q, Q) \operatorname{tr}(B[A, M]). \tag{32}$$

The six rotational tests therefore vanish exactly when $[A, B] = 0$. For $J_{ij} = x_i \partial_j - x_j \partial_i$, the coefficient is $-(2\,\mathrm{vol}(S^3)/3)\omega \mathcal{B}_{\mathrm{rad}}(Q, Q)[A, B]_{ij}$. This calculation concerns the restricted family (30), not every perturbation or every adjoint constraint mode.

6 Regular and singular quadrupole test states

For the finite quadrupole condition, consider the general map

$$V = \mathrm{Sym}_0(n, \mathbb{R}), \qquad \mu : V \times V \longrightarrow \mathfrak{so}(n), \qquad \mu(A, B) = [A, B], \qquad Z = \mu^{-1}(0), \qquad n \geq 2.$$

The metrics are the Frobenius metric on matrices and its product metric on $V \times V$. This is standard commuting symmetric-matrix geometry. Under the canonical identification $T^*V \simeq V \times V$, the commutator represents the moment map for simultaneous orthogonal conjugation, up to the sign fixed by the dual-pairing convention. The claims below do not concern the range of a gravitational constraint operator, interface conditions, or nonlinear spacetime continuation. The momentum-map context is classical [3].

Fix $(A, B) \in Z$. Simultaneous orthogonal diagonalization gives the *maximal* joint eigenspace decomposition

$$\mathbb{R}^n = \bigoplus_{s=1}^{r} E_s, \qquad A|_{E_s} = \alpha_s I, \qquad B|_{E_s} = \beta_s I, \qquad (\alpha_s, \beta_s) \neq (\alpha_t, \beta_t) \quad (s \neq t).$$

Write $m_s = \dim E_s$, let P_s be its orthogonal projector, and put $a_{st} = P_s a|_{E_t}$ and $a_{\mathrm{d}} = \sum_s P_s a P_s$, with analogous notation for b. In a simultaneous orthonormal eigenbasis, use repeated lists α_i, β_i and define

$$c_{ij} = \beta_j - \beta_i, \qquad d_{ij} = \alpha_i - \alpha_j, \qquad g_{ij}^2 = c_{ij}^2 + d_{ij}^2 \quad (i < j).$$

Proposition 6.1 (Rank, optimal inverse, and regularity)**.** *The derivative $L = D\mu_{(A,B)}$ is*

$$L(a,b) = [a,B] + [A,b], \qquad L(a,b)_{ij} = c_{ij}a_{ij} + d_{ij}b_{ij}.$$

Its nonzero singular values are exactly the positive g_{ij}. Equivalently, for each $s < t$ the singular value

$$\sqrt{(\alpha_s - \alpha_t)^2 + (\beta_s - \beta_t)^2}$$

occurs with multiplicity $m_s m_t$. Hence, with $d = \dim V = n(n+1)/2 - 1$ and $N = \dim \mathfrak{so}(n) = n(n-1)/2$,

$$\operatorname{rank} L = N - \sum_s \binom{m_s}{2}, \qquad \dim\ker L = 2d - N + \sum_s \binom{m_s}{2}.$$

The range comprises precisely the skew matrices S with $P_s S P_s = 0$ for all s. The Moore–Penrose inverse $R = L^\dagger$ is the pair of symmetric zero-diagonal matrices with

$$(RS)^{(1)}_{ij} = \frac{c_{ij}S_{ij}}{g_{ij}^2}, \qquad (RS)^{(2)}_{ij} = \frac{d_{ij}S_{ij}}{g_{ij}^2} \qquad (g_{ij} > 0),$$

and both entries zero when $g_{ij} = 0$. Here the two displayed components are extended symmetrically. Then LR is the orthogonal range projection. If at least one $g_{ij} > 0$,

$$\|R\|_{\mathfrak{so}(n)\to V\times V} = \frac{1}{\min_{g_{ij}>0} g_{ij}}.$$

For $L = 0$, $R = 0$ and $\|R\| = 0$. In particular, a right inverse on all of $\mathfrak{so}(n)$ exists exactly when every joint eigenvalue is simple. In that case R is the unique solution of minimum product norm for each target, and its operator norm is the smallest possible among all linear right inverses.

Proof. For $i < j$, the normalized symmetric and skew matrix units are respectively $(E_{ij} + E_{ji})/\sqrt{2}$ and $(E_{ij} - E_{ji})/\sqrt{2}$. On the two normalized input directions belonging to this edge, L is the single row (c_{ij}, d_{ij}). Distinct edges are orthogonal; all diagonal input directions are in the kernel. The trace restrictions affect only those diagonal inputs. Thus each edge contributes one singular value g_{ij} unless it lies within a joint eigenspace. This proves the singular-value, rank, and range statements. The inverse formula is the minimum-norm solution of the scalar equation $c_{ij}a_{ij} + d_{ij}b_{ij} = S_{ij}$. Consequently,

$$\|RS\|^2_{V\times V} = 2 \sum_{i<j:\, g_{ij}>0} \frac{S_{ij}^2}{g_{ij}^2}.$$

The norm formula follows by concentrating S on an edge of minimum positive gap. Every solution for that target has norm at least the minimum-norm solution, which also proves optimality among right inverses. There is no extra factor $\sqrt{2}$ in the singular values: the input and output Frobenius normalizations agree. □

Theorem 6.2 (All velocities of commuting curves)**.** *For $a, b \in V$, the following conditions are equivalent.*

1. *There is a C^1 curve $t \mapsto (A(t), B(t)) \in Z$ through (A, B) with derivative (a, b) at 0.*
2. *$L(a,b) = 0$ and $[a_{ss}, b_{ss}] = 0$ on every maximal joint eigenspace E_s.*
3. *$L(a,b) = 0$ and there are unrestricted $c, d \in V$ satisfying the second-order matrix equation*

$$L(c,d) = -2[a,b].$$

4. There is a real-analytic such curve with derivative (a, b) at 0.

For a velocity satisfying these conditions, define a skew matrix K by $K_{ss} = 0$ and, for $s \neq t$,

$$K_{st} = \frac{(\alpha_t - \alpha_s)a_{st} + (\beta_t - \beta_s)b_{st}}{(\alpha_t - \alpha_s)^2 + (\beta_t - \beta_s)^2}.$$

One explicit realization is

$$Q(t) = (I + tK/2)(I - tK/2)^{-1},$$
$$A(t) = Q(t)(A + ta_{\mathrm{d}})Q(t)^{\mathsf{T}}, \qquad B(t) = Q(t)(B + tb_{\mathrm{d}})Q(t)^{\mathsf{T}}.$$

This rational curve is defined and real analytic for every real t.

Proof. First, $L(a, b) = 0$ is necessary by differentiation. Between distinct joint blocks it reads

$$(\beta_t - \beta_s)a_{st} - (\alpha_t - \alpha_s)b_{st} = 0.$$

The coefficient pair is nonzero. Therefore the displayed definition of K gives

$$a_{st} = (\alpha_t - \alpha_s)K_{st}, \qquad b_{st} = (\beta_t - \beta_s)K_{st} \quad (s \neq t).$$

Symmetry of a, b and reversal of the gaps imply $K_{ts} = -K_{st}^{\mathsf{T}}$. In particular,

$$a_{st}b_{ts} - b_{st}a_{ts} = 0 \quad (s \neq t), \qquad P_s[a, b]P_s = [a_{ss}, b_{ss}]. \tag{33}$$

For necessity of the block condition, differentiability at 0 alone is enough. Put $X(t) = A(t) - A = ta + o(t)$ and $Y(t) = B(t) - B = tb + o(t)$. Every commutator with A or B has zero diagonal joint block, so the exact equation $[A + X(t), B + Y(t)] = 0$ gives

$$0 = P_s[X(t), Y(t)]P_s = t^2 P_s[a, b]P_s + o(t^2).$$

Divide by t^2 and use (33). This proves (1) $\Rightarrow$ (2) without assuming a second derivative of the curve.

For sufficiency, the block condition says $[a_{\mathrm{d}}, b_{\mathrm{d}}] = 0$. Both block-diagonal matrices commute with A and B, since the latter are scalar on each joint block. Thus $[A + ta_{\mathrm{d}}, B + tb_{\mathrm{d}}] = 0$ identically. For real t, $I - tK/2$ is invertible: if $(I - tK/2)x = 0$, taking the real inner product with x gives $\|x\|^2 = 0$. The Cayley matrix $Q(t)$ is orthogonal, has $Q(0) = I$, and has $Q'(0) = K$. Hence the proposed curve is symmetric, tracefree, real analytic, and commuting, and

$$A'(0) = a_{\mathrm{d}} + [K, A] = a, \qquad B'(0) = b_{\mathrm{d}} + [K, B] = b.$$

This proves (2) $\Rightarrow$ (4) $\Rightarrow$ (1).

Finally, Proposition 6.1 says the second-order equation is solvable exactly when $P_s[a, b]P_s = 0$ for all s. Under $L(a, b) = 0$, (33) identifies this with the block condition. This proves (2) $\Leftrightarrow$ (3). Explicit accelerations of the curve are

$$c = 2[K, a_{\mathrm{d}}] + [K, [K, A]], \qquad d = 2[K, b_{\mathrm{d}}] + [K, [K, B]],$$

and $R(-2[a, b])$ is the correction of minimum product norm. □

Remark 6.3. The condition is *not* $[a, b] = 0$ on all of $\mathbb{R}^n$. Off-block components of this commutator can be canceled by second-order corrections. For example, let $A = \mathrm{diag}(1, -1, 0, 0)$, $B = 0$, $a = E_{12} + E_{21}$, and $b = A$. Then $L(a, b) = 0$ and every diagonal joint block commutator is zero, although $[a, b] \neq 0$. Splitting a repeated maximal joint block into artificial one-dimensional blocks would instead discard genuine obstructions and is invalid.

Corollary 6.4 (Smooth points and a simple pencil)**.** *The following are equivalent at a commuting pair: L is surjective; all maximal joint eigenspaces are one dimensional; some real pencil $A+\tau B$ has simple spectrum; and Z is a C^1 embedded submanifold in a neighborhood of that pair. Its dimension there is*

$$2\dim V - \dim \mathfrak{so}(n) = \frac{n^2+3n-4}{2}.$$

Neither A nor B need individually have simple spectrum.

Proof. Surjectivity is Proposition 6.1, and the implicit-function theorem gives a smooth submanifold at every such point. If $m_s \geq 2$, choose an internal tracefree diagonal matrix a with entries $1, -1$ and $b = E_{ij} + E_{ji}$ on those two coordinates. The velocities $(a, 0)$ and $(0, b)$ are realized by straight commuting curves, while their sum fails Theorem 6.2. The velocity set of a C^1 embedded submanifold would be a vector space. Thus no repeated joint block is a smooth point. For distinct joint pairs, a collision $\alpha_i + \tau\beta_i = \alpha_j + \tau\beta_j$ excludes at most one value of τ per pair; if the two β values agree, no value is excluded. Avoiding this finite set gives a simple pencil. A repeated joint block prevents a simple pencil for every τ. □

A local polynomial chart at any regular pair can therefore be based on a simple pencil rather than on A alone. Fix a suitable τ and set $C = A + \tau B$, $D = B$. Every symmetric tracefree D commuting with a nearby simple C is uniquely

$$D = \sum_{k=1}^{n-1} c_k \left(C^k - \frac{\operatorname{tr}(C^k)}{n} I \right), \qquad A = C - \tau D, \quad B = D.$$

Interpolation on the n distinct eigenvalues proves existence and uniqueness. The coefficients depend smoothly on (C, D), for example by inverting the nonsingular Frobenius Gram matrix of the displayed polynomial basis. The $\dim V + n - 1$ parameters agree with Corollary 6.4. In dimension four, an example with neither individual matrix simple is

$$A = \operatorname{diag}(-1, -1, 1, 1), \qquad B = \operatorname{diag}(-1, 1, -1, 1).$$

All four joint pairs are distinct, and $A + 2B = \operatorname{diag}(-3, 1, -1, 3)$ is simple.

Dimension-four strata and the audited seeds. For $n = 4$, the complete multiplicity list gives

joint multiplicities	rank L	$\dim \ker L$
$1, 1, 1, 1$	6	12
$2, 1, 1$	5	13
$2, 2$	4	14
$3, 1$	3	15
4	0	18

Every row other than the first has a nonlinear set of realizable velocities. At the norm-two seed $A = B = \operatorname{diag}(1, -1, 0, 0)$, take $a = \operatorname{diag}(0, 0, 1, -1)$ and $b = E_{34} + E_{43}$. Their sum satisfies the linear equation but has $[a, b]_{34} = 2$ in the repeated block, so no C^1 commuting curve realizes that velocity. This proves the seed's genuine singularity, rather than merely a failure of the chosen equations to have independent derivatives.

For the norm-matched replacement

$$A = B = \frac{1}{\sqrt{10}} \operatorname{diag}(3, 1, -1, -3), \qquad \operatorname{tr}(A^2) = \operatorname{tr}(B^2) = 2,$$

the minimal single-matrix spectral gap is $\gamma = 2/\sqrt{10}$ and the minimal joint gap is $\sqrt{2}\gamma = 2/\sqrt{5}$. Thus the inverse bound is sharp:

$$\|R\| = \frac{\sqrt{5}}{2}.$$

When $A = B$ has distinct eigenvalues λ_i, the general inverse reduces to $a = -C/2$, $b = C/2$, where $C_{ij} = S_{ij}/(\lambda_i - \lambda_j)$ and $C_{ii} = 0$.

Scope for the quadrupole application. When the restricted charge equations have independently been established to reduce to $[A, B] = 0$ with a nonzero prefactor, the preceding statements classify their entire finite curve-velocity obstruction and the optimal linear inverse. They make no claim that charge-zero data, or finite second-order solvability, suffice for the Einstein–scalar bulk and shell equations. The linear map $D\mu$ and its range must not be substituted for the full constraint/interface operator.

7 Scope, reproducibility and next questions

The obstruction is a necessary condition for a specified classical exact family. It neither constructs a nonlinear family through a charge-zero state nor proves second-order solvability, stability or a positive physical energy. The matrix curve result is sufficient only for the finite equation $[A, B] = 0$. It cannot be substituted for solving the full Einstein–scalar constraint and interface operator. A simple matrix spectrum removes one finite-dimensional degeneracy; it does not supply a PDE right inverse.

The direct witness here remains away from the shell during the short slab. It therefore proves the obstruction without a contact theorem. The earlier technical checkpoint contains a separate finite-angular contact construction [11]; its stronger domain and compatibility hypotheses are not implicit in Lemma 4.1. The present paper does not infer a nonzero jet at the exact causal front.

A constructive nonlinear continuation would require identifying the range of the full linearized constraint and interface operator, the gauge and trace domains, and a second-order correction with all generated angular sectors. Further orders and estimates of the nonlinear remainder would then remain. The calculation makes no assertion about an observational scale or the origin of the Big Bang.

The accompanying source package contains the derivation, explicit finite matrix checks, deliberately incorrect-formula controls, input identities, and the primary-source comparison. The checks test algebra and selected exact examples. They are not a computer-certified analytic PDE proof or external peer review. Published earlier checkpoints are preserved as separate records; the present manuscript is a focused review draft. The draft and reproducible supplement are available in the public research repository.

AI assistance and author review. Substantial OpenAI Codex assistance was used in derivation, programming, literature comparison and drafting, including internal reviews by separately delegated GPT-6 Astra and GPT-6.1 Sol agents. These are AI-assisted checks, not independent human peer review. The human author is responsible for checking the mathematics, sources, interpretation and final disclosure before any submission. Funding, conflicts, contact details and all journal declarations must be confirmed by the author for the submitted version.

A Bulk reconstruction calculation

On $s > 0$, set $r = \sqrt{s}$, $\mathrm{d}y = \Omega \mathrm{d}r$, $\rho = r\Omega$, $p_b = \phi_y = 2rf/\Omega$ and $a = \rho_y/\rho$. Here p_b denotes the radial background scalar derivative, not the master velocity p of Section 4. The background equations are

$$(p_b)_y = U_\phi - 4ap_b, \quad a_y = -\rho^{-2} - p_b^2/3, \quad a^2 = \rho^{-2} + p_b^2/12 - U/6. \tag{34}$$

For $P = r^4 F$, define

$$\mathcal{M}[P] = P_{yy} - (2a + 2(p_b)_y/p_b)P_y + \rho^{-2}(\Box_{\mathrm{dS}_4} + 4)P - \tfrac{2}{3}p_b^2 P. \tag{35}$$

The ambient wave operator is $\Box_\eta = \partial_r^2 + 4r^{-1}\partial_r + r^{-2}\Box_{\mathrm{dS}_4}$. Differentiation gives $\mathcal{M}[r^4 F] = \Omega^{-2} r^4 \mathscr{L} F$. With $\Psi = P/\rho^2$ and $\chi = -3P_y/(\rho^2 p_b)$, the reconstructed metric is $h_{yy} = -4\Psi$, $h_{y\mu} = 0$, $h_{\mu\nu} = 2\rho^2 \Psi \gamma_{\mu\nu}$. For arbitrary spacetime functions Ψ, its Ricci variation is

$$\begin{aligned}
\delta R_{yy} &= -4\Psi_{yy} - 16a\Psi_y + 2\rho^{-2}\Box_\gamma \Psi, \\
\delta R_{y\mu} &= -3D_\mu(\Psi_y + 2a\Psi), \\
\delta R_{\mu\nu} &= \gamma_{\mu\nu}\left[-\Box_\gamma \Psi - \rho^2\{\Psi_{yy} + 10a\Psi_y + 6(a_y + 4a^2)\Psi\}\right].
\end{aligned} \tag{36}$$

Put $C = p_b\chi_y - (p_b)_y\chi - 3\rho^{-2}(\Box_\gamma + 4)\Psi + 2p_b^2\Psi$. Substitution gives

$$\delta\mathscr{E}_{y\mu} = 0, \quad \delta\mathscr{E}_{yy} = -2C/3, \quad \delta\mathscr{E}_{\mu\nu} = \rho^2\gamma_{\mu\nu}C/3, \quad C = -3\mathcal{M}[P]/\rho^2. \tag{37}$$

The scalar variation satisfies $p_b\delta\mathscr{S} = (\partial_y + 4a)C$, giving (14). The pre-contact support stays in this $s > 0$ patch, so no analytic continuation of an operator identity through $s = 0$ is used in the proof.

B Covariant consistency of the interface cancellation

This appendix records the local boundary calculation behind the same physical interpretation; the direct rotational proof does not depend on a general phase-space existence theorem. Use the Einstein–scalar bulk potential, with $q_{AB} = \delta g_{AB}$, $q = g^{AB}q_{AB}$ and $q^{AB} = g^{AC}g^{BD}q_{CD}$:

$$\theta^A = \sqrt{-g}\left[\tfrac{1}{2}(\nabla_B q^{AB} - \nabla^A q) - \nabla^A\phi\,\delta\phi\right]. \tag{38}$$

Represent the two outward boundary forms in the same co-oriented shell density convention, including the Noether potentials and corners. For $v_i^a = h^{ab}n_i^A\delta g_{i,Ab}$, let $\beta_i = -\frac{1}{2}i_{v_i}\epsilon_W$ and $\ell = (K_+ + K_- - \sigma)\epsilon_W$. The unrestricted GHY variation gives

$$\sum_i \theta_i|_W + \delta\ell = -\epsilon_W\left(\tfrac{1}{2}I^{ab}\delta h_{ab} + P\delta\varphi\right) + \mathrm{d}_W\beta, \qquad \beta = \beta_+ + \beta_-. \tag{39}$$

Normal and mixed metric variations have been retained. For a fixed common shell-tangent vector ξ, let $Q_{\xi,i} = -\frac{1}{2} *_i \mathrm{d}\xi_i^\flat$. Set $A_i^a = h^a{}_B n_i^A \nabla_A \xi_i^B$. Then

$$Q_{\xi,i}|_W = -\tfrac{1}{2}i_{A_i + K_i\cdot\xi}\epsilon_W, \qquad \beta_i(\mathcal{L}_\xi g) = -\tfrac{1}{2}i_{A_i - K_i\cdot\xi}\epsilon_W. \tag{40}$$

The arbitrary normal extension A_i cancels, leaving

$$B_\xi := \sum_i Q_{\xi,i}|_W + i_\xi\ell - \beta(\mathcal{L}_\xi) = -i_{I\cdot\xi}\epsilon_W. \tag{41}$$

The combination on the left is the established completed boundary charge [5, Eq. (2.51)]; its explicit Israel-residual value is the interface calculation here. Its vanishing uses the physical junction equation, not a fixed induced metric at an external wall.

For moving embeddings, exact pullback gives $\delta(Y^*g) = Y^*(\delta g + \mathcal{L}_\chi g)$, where $\chi = \delta Y \circ Y^{-1}$ [13]. Thus the second variation contains $\ddot{g} + 2\mathcal{L}_\chi \dot{g} + \mathcal{L}_{\dot{\chi}} g + \mathcal{L}_\chi^2 g$. Equation (2) concerns the full pullback, including these terms. The matching chart used in Section 3 retains them and the normalization of the shell normals. No vanishing of an unrestricted second-order graph has been assumed.

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